

OPEN LAB – PHASE 1

SMART TRAFFIC LIGHT CONTROL SYSTEM

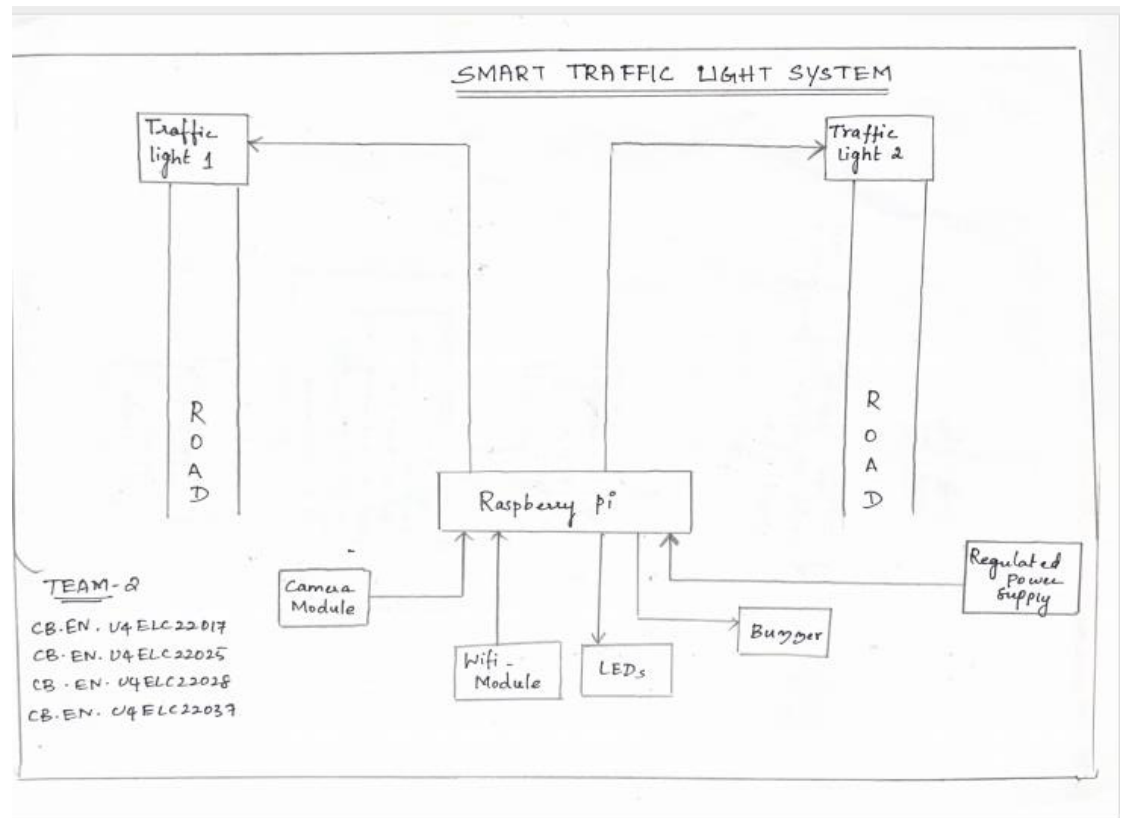
TEAM MEMBERS:

NO.	NAME	ROLL NUMBER
1	SUBA KEERTHANA K	CB.EN.U4ELC22017
2	LAKHSYA P	CB.EN.U4ELC22025
3	MITHILESHWARAN	CB.EN.U4ELC22028
4	NIVETHA GK	CB.EN.U4ELC22037

OBJECTIVE:

Traffic congestion and rule violations at urban intersections contribute to increased travel times, fuel consumption, and safety hazards. Traditional traffic light systems fail to dynamically adapt to varying traffic densities or prioritize emergency vehicles. This project aims to develop an intelligent traffic control system that integrates IoT, AI, and sensor technologies to monitor real-time traffic and dynamically adjust traffic signals. By leveraging tools like image processing, AI cameras, and sound sensor, the system seeks to optimize traffic flow, enhance pedestrian safety, and ensure compliance with traffic regulations.

CIRCUIT DIAGRAM



BUDGETING:

S. No	Component Required	Quantity	Amount	Justify the requirement
1	Raspberry Pi	1	6900	For handling image processing, sensor integration, and IoT tasks.

2	Camera Module	1	380	Camera for object detection and image processing (e.g., vehicle and pedestrian detection).
3	Sound Sensor Module	1	178	sensor used for detecting the ambulance.
4	Wifi module	1	650	Supports communication between the Raspberry Pi and other IoT devices (sensors, actuators, cameras) in the system.
8	Buzzer	1	15	Audio alerts and warnings in the traffic system.
6	5V Relay Module	1	50	Relay module to control traffic signal lights via Raspberry Pi.
7	Official Raspberry Pi Power Supply (5V, 3A)	1	583	Reliable power adapter for Raspberry Pi operation.
8	Connecting Wires	As required	-	Establish electrical connections between components.
9	5 mm LED Green-10 Red-4 Yellow-4	18	18	LEDs are used to simulate traffic lights at the intersection. The color-coded LEDs (green, red, yellow) represent the standard traffic signal colors for each lane.
10	Resistors 220 ohm	3	6	These resistors are used to limit the current passing through the LEDs, preventing damage and ensuring safe operation while maintaining uniform brightness levels for all traffic lights.
11	7 segment LED display	1	20	Used to indicate signal timing and provide clear information about the duration of green, yellow, or red lights to road users.

TIME SCHEDULE:

Week	Work Description	Name of the Lab in which work Involved
------	------------------	--

1	Initial Setup and Component Procurement	Embedded System laboratory
2	Image Processing for Traffic Density Estimation	Embedded System laboratory
3	Sound sensor integration	Embedded System laboratory
4	Emergency Vehicle Detection and Prioritization Logic	Embedded System laboratory
5	YOLOv3 Integration for Violation Detection	Embedded System laboratory
6	Traffic Signal Control Integration	Embedded System laboratory
7	Final Integration, Testing, and Optimization	Embedded System laboratory

LIST OF REFERENCES IN IEEE FORMAT:

- [1] V. Venkatesh, P. Raj, A. R. Anushiadevi, and K. A. Reddy, "An intelligent traffic management system based on the Internet of Things for detecting rule violations," *2023 International Conference on Advances in Computing, Communication and Applied Informatics (ACCAI)*, pp. 97-103, DOI: 10.1109/ACCAI58221.2023.10199293.
- [2] A. Pavani, P. Krishna Priya, M. Gupta, K. Joshi, B. Swathi, and M. Tiwari, "The use of images analysis for IoT-based smart traffic densities management," *2023 3rd International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE)*, pp. 1794-1798, DOI: 10.1109/ICACITE57410.2023.10182734.
- [3] C. Dema, Y. Wangchuk, L. B. Bhattarai, T. Ono, T. Gyelmo, and K. Muramatsu, "Traffic control system using AI camera and time of flight sensor," *2024 Third International Conference on Intelligent Techniques in Control, Optimization and Signal Processing (INCOS)*, DOI: 10.1109/INCOS59338.2024.10527746.

CONSOLIDATION OF RELEVANT INFORMATION FROM REFERENCE MATERIAL :

1. Traffic Rule Violation Detection (Paper 1: IoT and YOLO-based System)

- **Focus:**
 - Uses IoT and machine learning to address traffic rule violations (e.g., signal jumping, lane changes).
 - Incorporates YOLOv3 for real-time vehicle detection and classification.
 - Employs Python with OpenCV for image processing and TensorFlow for detection algorithms.
- **Key Features:**
 - Cameras installed near intersections capture vehicle behavior.
 - Immediate detection and reporting of violations to traffic servers.
 - Achieves ~93.83% accuracy in vehicle detection.
- **Relevance to Your Project:**
 - Incorporate real-time processing for traffic data to ensure immediate response and reporting.

2. IoT-Based Traffic Density Management (Paper 2: Image Processing with Raspberry Pi)

- **Focus:**
 - Manages traffic signals based on real-time traffic density estimation using image analysis.
 - Compares live traffic images with baseline images to compute vehicle density.
 - Uses Raspberry Pi for processing and IoT for remote monitoring.
- **Key Features:**
 - Traffic density is calculated by analyzing pixel differences in grayscale and binary images.
 - Results are transmitted via cloud platforms for real-time monitoring and control.
- **Relevance to Your Project:**
 - Utilize image comparison techniques for dynamic traffic signal adjustments based on density.
 - Incorporate IoT for centralized monitoring and remote control.

3. Traffic Control with AI Cameras (Paper 3: Pedestrian and Vehicle Priority)

- **Focus:**
 - Integrates AI cameras to prioritize pedestrian safety and manage vehicle traffic.
 - AI cameras detect pedestrian count.

- Uses Arduino Mega 2560 as the central controller for decision-making.
- **Key Features:**
 - Traffic and pedestrian lights are controlled dynamically based on pedestrian and vehicle data.
 - Data is relayed to cloud storage for logging and analysis.
- **Relevance to Your Project:**
 - Use AI cameras and ToF sensors for balancing pedestrian and vehicle priorities.
 - Implement dynamic signal control algorithms for real-time adjustments.